

Paper I - Basic Sciences in Urology (MCh Urology, Summer 2015)

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I-1. Describe in details basic patho-physiology of urine production and transport

Type: LAQ (25 marks)

Introduction

- Urine is formed by filtration, reabsorption, secretion and concentration, then propelled by peristalsis into a low-pressure bladder.

A. Renal blood flow (RBF)

- RBF is about **25% of cardiac output (CO)**; renal plasma flow (RPF) ~600 mL/min.
- Medullary flow is low because the vasa recta run in countercurrent. Medullary oxygen tension is only 10–20 mmHg, so the thick ascending limb and the S3 proximal segment work at the brink of hypoxia (vulnerable in acute tubular necrosis).
- **Autoregulation** keeps RBF and GFR (glomerular filtration rate) nearly constant at mean arterial pressure **80–180 mmHg**:
 - **Myogenic reflex**: afferent arteriolar smooth muscle contracts when stretched and dilates (prostaglandin-mediated) when pressure falls.
 - **Tubuloglomerular feedback (TGF)**: the macula densa senses tubular NaCl. Low NaCl releases renin, and angiotensin II constricts the efferent arteriole; high NaCl constricts the afferent arteriole.
- Non-steroidal anti-inflammatory drugs (NSAIDs) disable the myogenic arm; angiotensin blockade disables the efferent arm.

B. Glomerular filtration

- GFR is **~125 mL/min (~180 L/day)**; filtration fraction (FF = GFR/RPF) about **20%**.
- **Starling forces**: net filtration pressure $\approx$ glomerular hydrostatic 60 – Bowman's space 18 – plasma oncotic 32 = **~10 mmHg**.
- **Arteriolar rule**: afferent constriction lowers both RBF and GFR; efferent constriction lowers RBF but **raises** GFR.
- **Obstruction** raises Bowman's space pressure, so filtration falls.
- The barrier (fenestrated endothelium, basement membrane, podocytes) restricts molecules by size and negative charge, so albumin is excluded.

C. Tubular handling - segment-wise

Segment	Handling	Clinical link
Proximal convoluted tubule (PCT)	~ 65% of Na ⁺ and water (iso-osmotic); all glucose via SGLT2 (sodium-glucose cotransporter 2)	SGLT2 inhibitors
Thick ascending limb (TAL)	~ 25% of NaCl via NKCC2 (Na ⁺ -K ⁺ -2Cl ⁻ cotransporter); water-impermeable " diluting segment "	Loop diuretics
Distal convoluted tubule (DCT)	~5-7% of Na ⁺ ; Ca ²⁺ reabsorption	Thiazides lower urinary calcium
Collecting duct - principal cells	Na ⁺ /K ⁺ exchange (aldosterone); water via aquaporin-2 (AQP2)	Amiloride, spironolactone
Intercalated cells	Acid secretion	Distal renal tubular acidosis (RTA)

D. Concentration and dilution

- **Countercurrent multiplication:** the ascending limbs absorb NaCl without water, raising medullary osmolality from 300 to ~**1200 mOsm/kg**; urea recycling supplies about half of inner-medullary osmolality.
- The **vasa recta** act as countercurrent exchangers and do not wash out the gradient.
- **Antidiuretic hormone (ADH):** released when plasma osmolality rises or blood volume falls. Via the **V2 receptor** and cAMP (cyclic adenosine monophosphate) it inserts **AQP2** into the apical membrane of principal cells.
- Concentrated urine needs both a hypertonic medulla and ADH-driven AQP2.
- **Hormonal control:** angiotensin II and aldosterone retain Na⁺; atrial natriuretic peptide causes natriuresis.

E. Upper urinary tract transport

- **Pacemaker:** atypical smooth muscle cells in the **minor calyces/pelviccalyceal border**.
- Peristalsis is **myogenic**: the excised or denervated ureter keeps contracting. Nerves carry pain, not propulsion.
- **Electromechanical coupling:** calcium entry generates the action potential, which spreads through gap junctions; calcium-calmodulin activates myosin light-chain kinase.
- **Bolus propagation:** the ureter contracts behind each bolus with walls **coapted**; **2-6 waves/min**, ~20-80 cmH₂O (resting ~0-5 cmH₂O).
- **Effect of flow:** higher flow first increases frequency, then bolus size. At very high flow boluses **coalesce** and the ureter dilates **without obstruction** (relevant to diuretic renography).
- **Ureterovesical junction:** an intramural tunnel (**length:diameter** ≈ **5:1**, Paquin) prevents reflux. Sustained bladder pressure >~40 cmH₂O stops efflux and causes hydronephrosis.
- Endotoxin inhibits peristalsis; **α1-blockers** relax the distal ureter (medical expulsive therapy).
- The bladder stores ~400-500 mL at low pressure through compliance and sphincter tone, protecting upper tract drainage.

F. Pathophysiology of obstruction

Unilateral ureteral obstruction (UUO) - triphasic response:

Phase 1 (0–1.5 h):	RBF rises, ureteral pressure rises	(afferent dilatation)
Phase 2 (1.5–5 h):	RBF falls, pressure still rises	(efferent constriction)
Phase 3 (>5 h):	RBF falls, pressure falls	(afferent constriction, angiotensin II)

- GFR falls as RBF drops and back-pressure opposes filtration. Ischaemic injury begins **within 2 hours** of complete obstruction.
- **Tubular effects:** aquaporins (AQP1–4) and sodium transporters are down-regulated, causing a concentrating defect, salt wasting and **distal RTA**. Early obstruction wastes Na^+ , K^+ and bicarbonate; prolonged obstruction retains K^+ and H^+ .
- **Hypertension** (renin-mediated): ~77% with bilateral vs ~20% with unilateral obstruction.
- **Post-obstructive diuresis:** ~2% after unilateral and ~50% after bilateral relief, from AQP loss, washed-out medullary tonicity and retained solute.
- Chronic obstruction causes fibrosis via angiotensin II and transforming growth factor- β 1; recovery depends on duration and degree.

Diagram to draw: Nephron with segmental transporters and % Na^+ reabsorbed; triphasic UUO graph.

Recent advances

- SGLT2 inhibitors restore tubuloglomerular feedback; tolvaptan (V2 antagonist) slows polycystic kidney disease.

Key points to remember:

- RBF ~25% of CO; GFR 125 mL/min; FF 20%; net filtration pressure ~10 mmHg (60 – 18 – 32).
- PCT 65%, TAL 25%, DCT 5–7%; ADH: V2 → cAMP → AQP2.
- Pacemaker in minor calyces; peristalsis myogenic, 2–6/min.
- UUO triphasic; injury within 2 h; post-obstructive diuresis ~2% unilateral vs ~50% bilateral.

I-2. Describe female pelvic anatomy in relation to prolapse and SUI

Type: LAQ (25 marks)

Introduction

- Support depends on the **levator ani, endopelvic fascia/ligaments and bony pelvis** acting together. Failure causes **pelvic organ prolapse (POP)** and **stress urinary incontinence (SUI)**, effort leakage when urethral support or closure fails.

A. Bony pelvis

- Fixation points for repair, each with a nearby danger: **ischial spine** and **sacrospinous ligament** (pudendal vessels and nerve), **Cooper's ligament** (bleeding) and **sacral promontory** (left common iliac vein, right ureter).

B. Pelvic floor muscles (pelvic diaphragm)

Muscle	Attachment	Function
Pubococcygeus	Pubis to vagina, perineal body, anus	Closes the hiatus, lifts urethra and vagina
Puborectalis	Sling behind the anorectal junction	Anorectal angle
Iliococcygeus	Arcus tendineus levator ani to coccyx	Horizontal levator plate

- Tonic levator contraction keeps the **urogenital hiatus** closed; all POP passes through it.
- Muscle fails first, fascia second.** While the levator holds, ligaments carry little load. After birth injury or denervation they carry it all and eventually fail.
- On magnetic resonance imaging (MRI), up to **20% of primiparous women** have levator defects. Avulsion (in about a quarter of symptomatic women) predicts advanced, multicompartament prolapse.

C. Endopelvic fascia and ligaments

- Cardinal and uterosacral ligaments** suspend the apex; the uterosacral is the main apical anchor.
- Pubocervical fascia** supports the anterior wall and attaches laterally to the **arcus tendineus fasciae pelvis (ATFP, "white line")**, from the pubis to just above the ischial spine.
- Rectovaginal fascia** supports the posterior wall; **pubourethral ligaments** anchor the mid-urethra.

D. DeLancey's three levels of support (1992)

Level	Region	Structures	Defect causes
I - suspension	Cervix/upper third	Cardinal-uterosacral complex	Uterine/vault prolapse, enterocele
II - attachment	Middle third	ATFP, levator fascia	Cystocele, rectocele
III - fusion	Distal third	Perineal membrane, perineal body	Urethral hypermobility, perineal descent

- Central cystocele:** midline fascial weakness, **rugae lost**. **Paravaginal cystocele:** detachment from ATFP, **rugae preserved**.
- Posterior:** rectocele from rectovaginal fascia defect; perineal descent from perineal body failure.

- Level I is the keystone: apical loss underlies most anterior prolapse, so the apex must be corrected in any repair.

E. Urethral support and continence

- Continence zone is the **proximal and mid-urethra**. The oestrogen-sensitive vascular submucosa provides up to ~30% of closure; the striated sphincter adds tonic compression.

Theory	Mechanism	Implication
Pressure transmission (Enhörning 1961)	Hypermobility urethra descends out of the abdominal pressure zone	Restore position (Burch)
Hammock (DeLancey 1994)	Urethra compressed against a firm backing: anterior vaginal wall attached to ATRP and levator	Restore backing
Integral theory (Petros & Ulmsten 1990)	Pubourethral ligament, suburethral hammock and pubococcygeus at the mid-urethra	Tension-free mid-urethral sling
Intrinsic sphincter deficiency (ISD) (McGuire 1978)	Failure of urethral function itself	Sling, bulking

- ISD: Valsalva leak point pressure <60 cmH₂O or maximal urethral closure pressure <20 cmH₂O. Most SUI has some ISD, so slings work in hypermobility, whereas Burch does poorly in severe ISD.

F. Innervation

- **Pudendal nerve (S2-4):** sphincters; crosses the sacrospinous ligament near the ischial spine.
- **Nerve to levator ani (S3-5):** runs on the pelvic surface of the muscle. Electromyography shows levator neuropathy in about a quarter of primiparous women at 6 weeks.

G. Risk factors (Bump & Norton)

- **Predisposing:** genetics, connective tissue disorders (Ehlers-Danlos, Marfan).
- **Inciting: vaginal childbirth** (strongest; forceps worse), hysterectomy.
- **Promoting:** obesity, constipation, chronic cough.
- **Decompensating:** ageing; menopause is cited but not consistently associated.

H. POP-Q (Pelvic Organ Prolapse Quantification) staging (Bump 1996)

- **Hymen = 0;** points above negative, below positive (cm); tvl = total vaginal length.

Stage	Definition
0	No prolapse
I	Most distal point >1 cm above hymen
II	Within ± 1 cm of hymen
III	>1 cm below hymen but < (tvl - 2) cm
IV	Complete eversion ($\geq$ tvl - 2)

- Examination: cough stress test, repeated with prolapse reduced (occult SUI); Q-tip change >30° = hypermobility.

I. How anatomy guides surgery

Defect	Operation	Principle
SUI	Mid-urethral sling (MUS) : retropubic (Ulmsten 1995; tension-free vaginal tape, TVT) or transobturator (Delorme 2001)	Mid-urethral support
SUI	Burch colposuspension	Paravaginal fascia to Cooper's ligament
SUI – ISD	Autologous sling at bladder neck, bulking	Compression
Paravaginal cystocele	Paravaginal repair	Reattach to ATRP
Apical	Sacrocolpopexy (most durable)	Mesh to ligament over S1
Apical (vaginal)	Sacrospinous fixation 1.5–3 cm medial to ischial spine	Avoids pudendal bundle

- **April 2019:** the US Food and Drug Administration (FDA) stopped sale of transvaginal mesh for anterior POP; MUS and sacrocolpopexy mesh unaffected.

Diagram to draw: DeLancey levels I–III; hammock cross-section with a pressure arrow.

Recent advances

- 3D/4D translabial ultrasound for levator avulsion; mini-slings comparable to MUS (SIMS trial, 2022).

Key points to remember:

- DeLancey: I suspension, II attachment (ATRP), III fusion; level I is the keystone.
- Enhorning 1961; DeLancey hammock 1994; Petros & Ulmsten 1990; McGuire ISD 1978.
- Central cystocele rugae lost; paravaginal preserved. POP-Q stage II = ± 1 cm of hymen.
- Sacrospinous suture 1.5–3 cm medial to spine; FDA 2019: anterior transvaginal mesh withdrawn.

I-3a. Newer generation CT scan

Type: SAQ (10 marks)

Introduction

- Computed tomography (CT) is the reference imaging for stones, renal masses, haematuria and trauma. Hounsfield's first clinical scan was in 1971 (Nobel Prize 1979, with Cormack).

Evolution

Generation	Feature
Conventional (1971)	Step-and-shoot axial images
Helical/spiral CT (~1989)	Continuous table movement gives volumetric data
Multidetector CT (MDCT) (4-slice 1998 → 320-row)	Sub-millimetre slices, multiplanar and 3D reconstruction
Dual-source/dual-energy CT (2005)	Two energies; speed and material analysis
Photon-counting CT (PCCT) (FDA [US Food and Drug Administration] 2021)	Counts and energy-sorts each photon: no electronic noise, higher resolution, lower dose

Dual-energy CT (DECT)

- Acquires two energies (e.g. 80 and 140 kVp) and compares attenuation pixel by pixel.
- **Uric acid vs non-uric-acid stones** with near-perfect accuracy (chemolysis candidates).
- **Virtual unenhanced** images replace the true non-contrast phase; **iodine maps** separate hyperdense cysts from enhancing masses.
- Limitations: more dose than low-dose CT; limited field of view in obesity.

Low-dose CT KUB (kidney-ureter-bladder)

Protocol	Effective dose
Conventional CT abdomen/pelvis	10-20 mSv
Low-dose	<3-4 mSv
Ultra-low-dose	1-2 mSv (~1 mSv with deep learning)

- Cutting tube current keeps sensitivity ~95%. Use standard dose if body mass index >30.
- **EAU (European Association of Urology)**: ultrasound first, low-dose non-contrast CT if inconclusive. AUA (American Urological Association): CT first.

Reconstruction (dose reduction)

- **Filtered back projection**: fast but noisy, dose-hungry.
- **Iterative reconstruction**: roughly halves dose but slow, "waxy" images.
- **Deep learning reconstruction**: low noise at ultra-low dose, fast enough for emergencies.

CT urography (CTU)

- Triple-phase (unenhanced, nephrographic, excretory) gives ~20-30 mSv.
- **Split-bolus**: two injections, one combined nephrographic-excretory acquisition; less dose but may miss subtle urothelial lesions.

Diagram to draw: Timeline: axial → helical → MDCT → dual-source → PCCT.

Key points to remember:

- Doses: conventional 10–20, low-dose <3–4, ultra-low 1–2 mSv; 10 mSv $\approx$ 1 in 1000 cancer risk.
 - DECT: uric acid stones, virtual unenhanced, iodine maps.
 - Deep learning reconstruction enables $\sim$ 1 mSv CT.
 - PCCT FDA-cleared 2021.
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I-3b. History of Laparoscopy

Type: SAQ (10 marks)

Introduction

- Laparoscopy is endoscopic viewing of the peritoneal cavity through a pneumoperitoneum. It evolved from optics to diagnostic, therapeutic and finally robotic surgery.

Timeline

Year	Pioneer	Milestone
1806	Bozzini	Lichtleiter (candle-lit light conductor)
1877	Nitze	Cystoscope with distal electric light and lenses
1901	Kelling	Viewed the air-insufflated abdomen of a dog
1910	Jacobaeus	First human series; coined "laparoscopy"
1924	Zollikofer	CO ₂ insufflation
1938	Veress	Spring-loaded needle
1960s	Hopkins	Rod-lens system (bright, narrow scopes)
1970s-80s	Semm	Automatic insufflator; laparoscopic appendectomy (1980)
1970s	Hasson	Open access technique
1976	Cortesi	Impalpable testis – first urological use
1987	Mouret	Video-laparoscopic cholecystectomy
1990	Clayman	First laparoscopic nephrectomy (published 1991)
1992	Gaur (India)	Balloon dilatation of the retroperitoneum
1997	Schuessler	First laparoscopic radical prostatectomy reported
1995	Ratner	Laparoscopic live-donor nephrectomy
2000	Guillonneau & Vallancien	Standardised laparoscopic prostatectomy
2000	da Vinci	FDA (US Food and Drug Administration) clearance
2000	Binder	First robot-assisted radical prostatectomy (RARP)
Early 2000s	Menon	Vattikuti Institute Prostatectomy (VIP) standardised RARP
2007	Raman; Rane	Single-port nephrectomy
2018	da Vinci SP	Single-port robot

Phases of evolution

Optics → Diagnostic → Therapeutic (Semm) → Video laparoscopy
→ Urological surgery → Robotic → Single-site / telesurgery

Why the robot spread

- Laparoscopic prostatectomy stalled because intracorporeal suturing was hard. The robot gave wristed instruments and a 3D magnified view, removing that barrier.
- da Vinci generations: S (2006, fourth arm), Si (2009, dual console), Xi (2014, flexible docking), SP (2018).

Indian contribution

- Gaur's balloon made retroperitoneoscopy practical. India has an indigenous robot (**SSI Mantra**) with telesurgery demonstrations.

Recent advances

- New multi-port robots (Versius, Hugo RAS, Hinotori), long-distance telesurgery and haptic feedback.

Diagram to draw: Timeline arrow 1806 → 1901 → 1910 → 1938 → 1990 → 2000.

Key points to remember:

- Bozzini 1806; Kelling 1901 (dog); Jacobaeus 1910 (human); Veress 1938.
 - Hopkins rod-lens; Semm – father of operative laparoscopy.
 - Urology: Cortesi 1976; Clayman 1990; Gaur balloon 1992; Ratner 1995.
 - da Vinci FDA 2000; Binder 2000 first RARP; Menon – VIP.
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I-3c. Renin-angiotensin system

Type: SAQ (10 marks)

Definition

- Hormonal cascade regulating **blood pressure, extracellular volume, Na⁺/K⁺ balance and glomerular filtration rate (GFR)**.

Cascade

Angiotensinogen (liver)
↓ renin (juxtaglomerular cells) – rate-limiting step
Angiotensin I
↓ angiotensin-converting enzyme (ACE), mainly in lung
Angiotensin II → AT1 and AT2 receptors
↓ ACE2
Angiotensin (1-7) → Mas receptor (vasodilator, anti-fibrotic)

Juxtaglomerular (JG) apparatus

- **JG (granular) cells** in the afferent arteriole wall secrete renin and act as baroreceptors.
- The **macula densa** senses tubular NaCl; extraglomerular mesangial cells complete the apparatus.

Control of renin release

- **Stimulated by** low perfusion pressure, low NaCl at the macula densa, **sympathetic β1 activity** and hypovolaemia.
- **Inhibited by** angiotensin II (negative feedback), atrial natriuretic peptide and β-blockers.

Receptors and actions

- **AT1 (angiotensin II type 1) receptor:** vasoconstriction, **aldosterone** release (distal Na⁺ retention, K⁺ loss), direct proximal Na⁺ reabsorption, thirst, ADH (antidiuretic hormone) release and **fibrosis**.
- **AT2 receptor:** vasodilatation; important in fetal urinary tract development.
- The **ACE2-angiotensin (1-7)** arm is protective.

Intrarenal effects on GFR

- Angiotensin II constricts the **efferent more than the afferent** arteriole, maintaining GFR when perfusion is low.
- Therefore **ACE inhibitors and angiotensin receptor blockers (ARBs) can cause an acute fall in GFR** in bilateral renal artery stenosis, stenosis of a solitary kidney or volume depletion.

Urological relevance

- **Renovascular hypertension (Goldblatt kidney):** atherosclerosis or fibromuscular dysplasia. A **renal vein renin ratio >1.5** lateralises and predicts response to revascularisation or nephrectomy.
- **Page kidney:** subcapsular haematoma compresses the parenchyma, causing renin-mediated hypertension; ACE inhibitor/ARB, drainage if refractory.
- **Reninoma (JG cell tumour):** young women, small (≤3 cm) mass, refractory hypertension and **hypokalaemia**; excision (partial nephrectomy) cures most.
- **Drugs:** ACE inhibitors, ARBs and aliskiren are contraindicated in pregnancy.

Diagram to draw: JG apparatus and the cascade flowchart.

Recent advances

- Renal denervation (FDA [US Food and Drug Administration] approval 2023) for resistant hypertension.

Key points to remember:

- Renin (JG cells) is rate-limiting; released by low pressure, low NaCl, β_1 .
 - Efferent constriction maintains GFR → ACE inhibitor risk in bilateral stenosis.
 - AT1 = pressor/fibrotic; AT2 and ACE2-Ang(1-7) = protective.
 - Renin ratio >1.5 lateralises; reninoma → excision cures most.
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I-3d. Patho-physiology of Brain stem death

Type: SAQ (10 marks)

Definition

- **Irreversible loss of the capacity for consciousness and of the capacity to breathe** from permanent brainstem failure; legally death. It enables donation after brain death (DBD), with minimal warm ischaemia.

Pathophysiology

Brain injury → oedema → raised intracranial pressure (ICP)
→ falling cerebral perfusion → herniation ("coning")
→ Pons: Cushing reflex (hypertension, bradycardia)
→ Medulla: catecholamine storm (hypertension, tachyarrhythmia)
→ Loss of sympathetic tone → vasodilatation, hypotension

- The storm causes myocardial injury and neurogenic pulmonary oedema.
- **Endocrine:** loss of posterior pituitary antidiuretic hormone causes **diabetes insipidus (DI)** with polyuria and hypernatraemia. Thyroid hormone and cortisol fall; hyperglycaemia is common.
- **Hypothermia** from loss of hypothalamic control; necrotic brain releases thromboplastin, causing **disseminated intravascular coagulation (DIC)**.
- A systemic inflammatory response injures donor organs.

Diagnosis

- **Preconditions:** apnoeic ventilated coma with a known irreversible cause. **Exclude** sedatives, neuromuscular blockers, hypothermia (thresholds about 35–36 °C depending on protocol) and metabolic or endocrine disturbance.
- **All brainstem reflexes absent:** pupillary, corneal, vestibulo-ocular (~50 mL ice-cold water), oculocephalic, cranial motor response to pain, gag and cough.
- **Apnoea test:** pre-oxygenate, disconnect the ventilator. No breathing effort when **arterial CO₂ (PaCO₂) reaches ≥60 mmHg** confirms apnoea.

Indian legal framework

- Recognised by the **Transplantation of Human Organs Act (THOA) 1994**, amended 2011, Rules 2014.
- Certified by a **4-member board:** hospital in-charge, an independent specialist, a neurologist/neurosurgeon and the treating doctor; no transplant team member.
- **Two sets of tests at least 6 hours apart.**

Donor management

- **Rule of 100s:** systolic blood pressure >100 mmHg, urine >100 mL/h, PaO₂ (arterial oxygen tension) >100 mmHg, haemoglobin >100 g/L.
- **Vasopressin** is preferred because it also treats DI; add thyroid hormone, methylprednisolone and insulin as needed.

Diagram to draw: Herniation sequence leading to storm, DI, hypothermia and DIC.

Recent advances

- Hypothermic machine perfusion of DBD kidneys reduces delayed graft function.

Key points to remember:

- Raised ICP → Cushing reflex → storm → hypotension, DI, hypothermia, DIC.
 - Exclude drugs, hypothermia, metabolic causes.
 - Apnoea test: no breathing at $\text{PaCO}_2 \geq 60$ mmHg.
 - THOA 1994: 4-member board, two tests ≥ 6 h apart; Rule of 100s.
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I-3e. Male reproductive immunology

Type: SAQ (10 marks)

Introduction

- Haploid germ cells first appear at puberty, after self-tolerance is established, so their antigens are "foreign". The testis is therefore an **immune-privileged** site.

Mechanisms of immune privilege

1. Anatomical barrier - blood-testis barrier (BTB)

- Formed at **puberty** by **tight junctions between Sertoli cells**, the strongest intercellular barrier in the body.
- It divides a **basal compartment** (spermatogonia, plasma-like fluid) from an **adluminal compartment** (later spermatocytes, spermatids).

2. Functional (immunological) barrier

- A small continuous leak of sperm antigen at weak points (**rete testis, efferent ductules, epididymis**) induces suppressor T cells and tolerance; a large antigen load provokes a true immune response.
- **Sertoli cell mediators:** galectin-1, transforming growth factor- β (TGF- β), interferon- γ , soluble Fc receptor; local **androgens** are immunosuppressive.
- Testicular insult before puberty does not induce antisperm antibodies; the barrier also limits chemotherapy access (isolated testicular relapse).

Antisperm antibodies (ASA)

Causes

- **Obstruction** (vasectomy commonest), torsion, trauma, biopsy, infection (orchitis, prostatitis), heat (varicocele, cryptorchidism), genetic (human leukocyte antigen HLA-B28).

Effects

- Agglutination, poor motility, failure to penetrate cervical mucus and to bind the zona. Significance depends on binding to **sperm heads**.

Testing

- **Mixed antiglobulin reaction (MAR)** on fresh semen or **immunobead test** on washed sperm. **$\geq 50\%$ of motile sperm bound** is significant (WHO [World Health Organization] 2010).

Treatment

- Corticosteroids: two controlled trials conflict, and they risk avascular necrosis.
- Sperm washing has little evidence, because it removes only unbound antibody.
- **IVF (in vitro fertilisation)/ICSI (intracytoplasmic sperm injection)** is now the usual treatment.

Inflammation and infection

- **Leukocytospermia:** **$>1 \times 10^6$ peroxidase-positive leukocytes/mL**. Reactive oxygen species and cytokines damage sperm; treat infection if found.
- **Orchitis:** post-pubertal mumps (atrophy), bacterial, tuberculous or autoimmune.

Diagram to draw: Seminiferous tubule with Sertoli tight junctions dividing basal and adluminal compartments.

Recent advances

- Sperm DNA fragmentation testing in men with ASA or leukocytospermia.

Key points to remember:

- BTB = Sertoli-Sertoli tight junctions; basal vs adluminal; formed at puberty.
 - Privilege is also active: antigen leak at rete/epididymis → tolerance; galectin-1, TGF- β , androgens.
 - ASA: vasectomy commonest; $\geq 50\%$ bound significant; IVF/ICSI preferred.
 - Leukocytospermia >1 million/mL.
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I-3f. Principles of antimicrobial therapy in UTI

Type: SAQ (10 marks)

Classification

- **Uncomplicated** urinary tract infection (UTI): healthy non-pregnant women, normal tract.
- **Complicated**: men, pregnancy, obstruction, stones, catheter, diabetes, immunosuppression.
- **Recurrent**: ≥ 2 infections in 6 months or ≥ 3 in 12 months.

Principles

- Treat the patient, not the culture; culture all except simple cystitis in women.
- **Site decides the level that matters**: urine level in cystitis; tissue and serum levels in pyelonephritis, prostatitis and bacteraemia.
- Nitrofurantoin and fosfomycin: **high urine, no tissue levels**, so **cystitis only**.
- **Pharmacodynamics**: β -lactams are **time-dependent**; aminoglycosides and fluoroquinolones (FQs) are **concentration-dependent**, so once-daily gentamicin maximises peak:MIC (minimum inhibitory concentration).
- **Resistance thresholds**: trimethoprim-sulfamethoxazole (TMP-SMX) if local resistance $\leq 20\%$; oral FQ for pyelonephritis if $\leq 10\%$.
- **Collateral damage**: FQs and cephalosporins select resistance and *C. difficile*; avoid in simple cystitis.
- Shortest effective course; **drain** an obstructed infected kidney.

Empirical choices

Scenario	First-line	Duration
Cystitis (women)	Nitrofurantoin; fosfomycin 3 g ; pivmecillinam	5 d ; single dose; 3–5 d
Alternative	TMP-SMX 160/800 mg twice daily	3 d
Cystitis in men	TMP-SMX or per culture	7 d (= 14 d; Drekonja 2021)
Pyelonephritis (oral)	Ciprofloxacin, or levofloxacin 750 mg	7 d; 5 d
Complicated UTI	Aminoglycoside $\pm$ ampicillin, or 3rd-generation cephalosporin	7–14 d
Catheter-associated	Change catheter if in >2 weeks	7 d
Urosepsis	Broad-spectrum within 1 h + source control	By response

Asymptomatic bacteriuria (ASB)

- Treat **only in pregnancy and before mucosa-breaching urological procedures**.

Prophylaxis

- Recurrent UTI: vaginal oestrogen, **methenamine hippurate** (non-inferior to antibiotics; ALTAR 2022), post-coital/low-dose antibiotics.
- Surgical: single dose within 1 hour of incision; ≤ 24 hours for PCNL (percutaneous nephrolithotomy) and prostheses.

Diagram to draw: Concentration-time curve showing peak:MIC and time above MIC.

Recent advances

- New oral agents for uncomplicated UTI: pivmecillinam (US approval 2024) and gepotidacin (2025).

Key points to remember:

- Recurrent = ≥ 2 in 6 months or ≥ 3 in 12 months.
 - Cystitis: nitrofurantoin 5 d, fosfomycin 3 g single, TMP-SMX 3 d (if resistance $\leq 20\%$).
 - Pyelonephritis: ciprofloxacin 7 d or levofloxacin 750 mg 5 d (if FQ resistance $\leq 10\%$).
 - ASB: treat only in pregnancy/before mucosal surgery.
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Rapid Revision Sheet - Paper I

All the key points from this paper in one place. Read this in the last 30 minutes before the exam.

I-1. Describe in details basic patho-physiology of urine production and transport (LAQ)

- RBF ~25% of CO; GFR 125 mL/min; FF 20%; net filtration pressure ~10 mmHg (60 - 18 - 32).
- PCT 65%, TAL 25%, DCT 5-7%; ADH: V2 → cAMP → AQP2.
- Pacemaker in minor calyces; peristalsis myogenic, 2-6/min.
- UUU triphasic; injury within 2 h; post-obstructive diuresis ~2% unilateral vs ~50% bilateral.

I-2. Describe female pelvic anatomy in relation to prolapse and SUI (LAQ)

- DeLancey: I suspension, II attachment (ATFP), III fusion; level I is the keystone.
- Enhorning 1961; DeLancey hammock 1994; Petros & Ulmsten 1990; McGuire ISD 1978.
- Central cystocele rugae lost; paravaginal preserved. POP-Q stage II = ± 1 cm of hymen.
- Sacrospinous suture 1.5-3 cm medial to spine; FDA 2019: anterior transvaginal mesh withdrawn.

I-3a. Newer generation CT scan (SAQ)

- Doses: conventional 10-20, low-dose <3-4, ultra-low 1-2 mSv; 10 mSv $\approx$ 1 in 1000 cancer risk.
- DECT: uric acid stones, virtual unenhanced, iodine maps.
- Deep learning reconstruction enables ~1 mSv CT.
- PCCT FDA-cleared 2021.

I-3b. History of Laparoscopy (SAQ)

- Bozzini 1806; Kelling 1901 (dog); Jacobaeus 1910 (human); Veress 1938.
- Hopkins rod-lens; Semm - father of operative laparoscopy.
- Urology: Cortesi 1976; Clayman 1990; Gaur balloon 1992; Ratner 1995.
- da Vinci FDA 2000; Binder 2000 first RARP; Menon - VIP.

I-3c. Renin-angiotensin system (SAQ)

- Renin (JG cells) is rate-limiting; released by low pressure, low NaCl, β_1 .
- Efferent constriction maintains GFR → ACE inhibitor risk in bilateral stenosis.
- AT1 = pressor/fibrotic; AT2 and ACE2-Ang(1-7) = protective.
- Renin ratio >1.5 lateralises; reninoma → excision cures most.

I-3d. Patho-physiology of Brain stem death (SAQ)

- Raised ICP → Cushing reflex → storm → hypotension, DI, hypothermia, DIC.
- Exclude drugs, hypothermia, metabolic causes.
- Apnoea test: no breathing at $\text{PaCO}_2 \geq 60$ mmHg.
- THOA 1994: 4-member board, two tests ≥ 6 h apart; Rule of 100s.

I-3e. Male reproductive immunology (SAQ)

- BTB = Sertoli-Sertoli tight junctions; basal vs adluminal; formed at puberty.
- Privilege is also active: antigen leak at rete/epididymis → tolerance; galectin-1, TGF- β , androgens.
- ASA: vasectomy commonest; $\geq 50\%$ bound significant; IVF/ICSI preferred.
- Leukocytospermia >1 million/mL.

I-3f. Principles of antimicrobial therapy in UTI (SAQ)

- Recurrent = ≥ 2 in 6 months or ≥ 3 in 12 months.
- Cystitis: nitrofurantoin 5 d, fosfomycin 3 g single, TMP-SMX 3 d (if resistance $\leq 20\%$).
- Pyelonephritis: ciprofloxacin 7 d or levofloxacin 750 mg 5 d (if FQ resistance $\leq 10\%$).
- ASB: treat only in pregnancy/before mucosal surgery.